

Confidence scoring for deep learning-predicted antibody–antigen complexes: AntiConf as a precision-driven metric

Serbülent Ünsal^{1,2}, Benjamin Holland¹, Inci Sardag³, Emel Timucin^{1,4,*}

¹Antiverse, Antiverse Ltd., sbarc/spark, Maindy Road, CF24 4HQ Cardiff, United Kingdom

²Faculty of Medicine, Karadeniz Technical University, Kanuni Campus, 61080 Trabzon, Turkey

³Department of Molecular Biology and Genetics, Bogazici University, North Campus, 34342 Istanbul, Turkey

⁴Department of Molecular Biology and Genetics, Gebze Technical University, Cumhuriyet 2254 St., Gebze, 41400 Kocaeli, Turkey

*Corresponding author. Department of Molecular Biology and Genetics, Gebze Technical University, Cumhuriyet 2254 St., Gebze, 41400 Kocaeli, Turkey. E-mail: emelt@gtu.edu.tr

Abstract

Accurate determination of antibody–antigen (Ab–Ag) complex structures is critical for therapeutic development. While deep learning-based methods, beginning with AlphaFold2 (AF2), have revolutionized multimer predictions, the optimal strategies for Ab–Ag modeling, and the reliability of their confidence scores remain active areas of research. This study evaluates the performance of AF2, Boltz-1, Boltz-1x, Boltz-2, Chai-1, Protenix, Protenix-1, OpenFold3, and ESMFold, on a curated dataset of 200 Ab–Ag complexes. Among the nine methods tested, Protenix-1 emerged as the top performer, with Chai-1 consistently ranking second across multiple success metrics, closely followed by AF2. We observed diverse effects of recycling iterations, with AF2, Chai-1, and Protenix variants benefiting from increased cycles, unlike Boltz variants. We analyzed various model confidence scores, noting high precision from pDockQ2 and high recall from predicted Template-Modeling (pTM) score. By integrating these two scores, we developed antibody confidence (AntiConf), a novel metric that achieves superior performance for all methods in terms of precision and recall. These strengths make AntiConf a valuable post score for both computational predictions and downstream experimental workflows, reflecting its potential to improve Ab–Ag complex predictions by AF2 and AF3 architectures. Altogether, this study addresses current limitations in deep learning-based Ab–Ag complex prediction, showcasing the potential of AntiConf for future assessment studies, and providing a guideline for improving the accuracy of Ab–Ag complex prediction.

Keywords Alphafold2, AF3-based implementations, multimer prediction, antibody–antigen complexes, model confidence scores

Introduction

Experimental determination of antibody–antigen (Ab–Ag) complexes is a resource-intensive process, underscoring the critical need for computational methods as efficient and scalable alternatives [1–3]. The landscape of protein structure prediction has evolved rapidly with the release of AlphaFold2 (AF2), which demonstrated striking performance in modeling complex biomolecular interactions, including Ab–Ag complexes [4–6]. Notably, AlphaFold3 (AF3), which introduces a diffusion-based architecture, has demonstrated improved accuracy over AF2 [7] for Ab–Ag complexes [8–11]. However, the AF3 model and the limitations of its webserver implementation currently restrict its utility for large-scale, automated pipelines required in high-throughput analysis. This has motivated a new generation of open-source models that replicate the architectural breakthroughs of AF3 while addressing specific limitations of the original framework. Prominent examples include Boltz-1 [12], Boltz-2 [13], Chai-1 [14], Protenix [15], Protenix-1 [16], HelixFold3 [17], and OpenFold3 [7, 18].

Among these, the recently introduced Protenix-1 has been shown to improve upon AF3 in accuracy on Ab–Ag predictions [16]. Despite this growing landscape of tools, independent benchmarking efforts on Ab–Ag complexes [8–11] have been limited in the number of methods evaluated and/or dataset size, or have not systematically examined practical factors such as recycling iterations that affect prediction quality. Moreover, standard model confidence scores often fail to reliably distinguish accurate from inaccurate predictions, limiting their utility for prioritizing candidates for experimental validation in antibody therapeutic development.

This study addressed these gaps by providing a comprehensive benchmark that extends prior efforts in both method coverage and dataset size. Here, we evaluated nine distinct methods, including recently released tools (Boltz-1x, Protenix, and Protenix-1) on a 200 Ab–Ag complex set. Overall, our analysis revealed three AF3-based implementations, Protenix-1, Chai-1, and OpenFold3 along with AF2 as the best performers distinct strengths in both Ab and Nb modeling.

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Our analysis further highlighted the importance of specific protocols such as recycling and sampling and reveals that standard confidence metrics frequently misclassify model quality. To resolve this, we developed antibody confidence (AntiConf), a composite metric that provides a superior balance of precision and recall compared with existing approaches, allowing us to define robust thresholds for selecting high-confidence models. Overall, our benchmarking supports a pragmatic, multi-model strategy for prospective end-to-end modeling pipelines starting with sequences.

Materials and methods

Dataset

We assembled a dataset of 200 Ab–Ag complexes from two recent studies [5, 19]. The first subset, consisting of 39 Protein Data Bank (PDB) complexes from Yin and Pierce [5], was directly incorporated. The second subset was refined from the initial 210 complex dataset from Hitawala and Gray [19], after removing unbound structures and complexes without proteinaceous antigens. Entries representing alternative biological units of the same target were also removed, retaining the first assembly. Sequences were reconstructed using PDB SEQRES records and/or UniProt entries. The final dataset was composed of 200 Ab–Ag complexes (124 Abs and 76 Nbs) (Fig. S1).

Model generation and evaluation

AF2 (v2.3.2) [4] was implemented through ColabFold [6, 20] using MMseqs2 [21] and the UniRef+Environmental database [22]. A total of 25 predictions per complex were generated for all methods, except for ESMFold [23], for which a single model was generated. A five-seed approach was used for AF2, while for AF3-based implementations of Boltz-1 (v0.4.0) [12], Boltz-1x (v1.0.0) (Boltz-steering) [12], Boltz-2 (v2.0.3) [13], Chai-1 [14], Protenix (v0.5.0) [15], Protenix-1 (v1.0.0) [16], and OpenFold3 [7, 18] were run using a single seed. Before assembling the final model, different number of recycling iterations were tested for all methods except for ESMFold and OpenFold3. All predictions were conducted via Tamarind Bio [24]. Model accuracy was assessed by DockQ v2 [25] and Critical Assessment of Prediction of Interactions (CAPRI) system [26]. DockQ scores and thresholds aligned well with the respective CAPRI categories (Fig. S2).

Calculation of confidence scores

For each chain, a residue was classified as an interface residue if any of its intermolecular $C\alpha$ – $C\alpha$ distances is ≤ 8 Å. The pLDDT scores and predicted aligned error (PAE) scores were extracted from the model files or global PAE matrix for the interface residues. PAE scores were normalized according to Zhu *et al.* [27]. The extracted ipLDDT and iPAE scores were averaged for either the antigen alone or for both interacting chains.

We calculated the pDockQ2 score, which combines ipLDDT and normalized iPAE. Specifically, the product of ipLDDT and normalized iPAE scores was fitted to a four-parameter sigmoid function to obtain the pDockQ2 score [27]:

$$\text{pDockQ2}(x) = \frac{L}{1 + \exp(-k(x - x_0))} + b$$

where x represents the product of ipLDDT and normalized iPAE. The curve parameters were adopted from Zhu *et al.* [27] as follows: $L = 1.3103$, $x_0 = 84.7326$, $k = 0.0747$, and $b = 0.0050$.

A composite score was computed as a weighted combination of pTM and pDockQ2 scores as follows:

$$\text{AntiConf} = 0.3\text{pDockQ2} + 0.7\text{pTM}$$

wherein the weights were derived via logistic regression on AF2 predictions. We also performed a grid search optimization across the entire dataset, resulting in nearly identical weights (0.67/0.33) with a global AUC of 0.90.

Results and discussion

Performance assessment based on CAPRI classification

We compiled a dataset of 200 Ab–Ag complexes, building upon two prior datasets released after the AF2 training cutoff (30 September 2021) [5, 8]. AF2 (v2.3.2), Boltz-1, Boltz-1x, Chai-1, and ESMFold were run on the full dataset of 200 Ab–Ag complexes. Due to memory constraints imposed by their architectures, Protenix, Protenix-1, and OpenFold3 could not process all models, resulting in evaluation sets of 174, 191, and 181 complexes, respectively. Additionally, Boltz-2, which uses a more recent training cutoff (2023), was evaluated on a subset of 18 complexes deposited after its training data.

To address potential data leakage, we compared the complementarity determining region (CDR) sequences in our dataset against PDB complexes deposited before the training cutoff using SabDab [28], identifying 36 complexes (31 Abs and 5 Nbs) with 70%–82% CDR sequence identity to training data. Evaluations performed with and without these complexes showed no inflation or significant change in prediction success (Fig. S3), so we retained the complete dataset for all subsequent analyses.

Figure 1 summarizes (left) the overall success based on the number of unique targets achieving at least CAPRI-acceptable accuracy, and (right) the distribution of models across distinct CAPRI categories. While quantifying the number of high/medium-quality models is valuable for understanding the methods' ability to produce refined interfaces [5, 19], success based on counts within each CAPRI category carries the risk of overestimating the performance. In line with this, the success rate calculated as the sum of unique counts in each CAPRI category (Fig. 1, right) is higher than unique counts with at least acceptable accuracy (Fig. 1, left). For example, AF2, when accounting for each CAPRI category individually, correctly predicted 122 targets (44 acceptable, 36 medium, and 42 high quality); however, only 88 of these were unique. Thus, we based our ultimate comparison of the method's predictive success on the rate calculated as the number of unique targets predicted with at least acceptable accuracy (Fig. 1, left).

For the complete dataset, Protenix-1 achieved the highest success rate with 88 unique targets (46%, 88/191), closely followed by AF2 (44%, 88/200) and Chai-1 (41%, 82/200) (Fig. 1, top row). Among these top performers, AF2 produced the largest fraction of high-quality models (42), demonstrating its capacity for capturing refined interfaces. Notably, despite being evaluated on a significantly smaller subset of targets, Boltz-2 showed promising performance, yielding exclusively high- and medium-quality models. Protenix-1 and Chai-1 were the second and third best methods for generating high-quality models, respectively. Additionally, Chai-1 and Protenix-1 stood out for producing the highest number of medium-quality models. Among all tested methods, OpenFold3 produced the fewer number of high accuracy models.

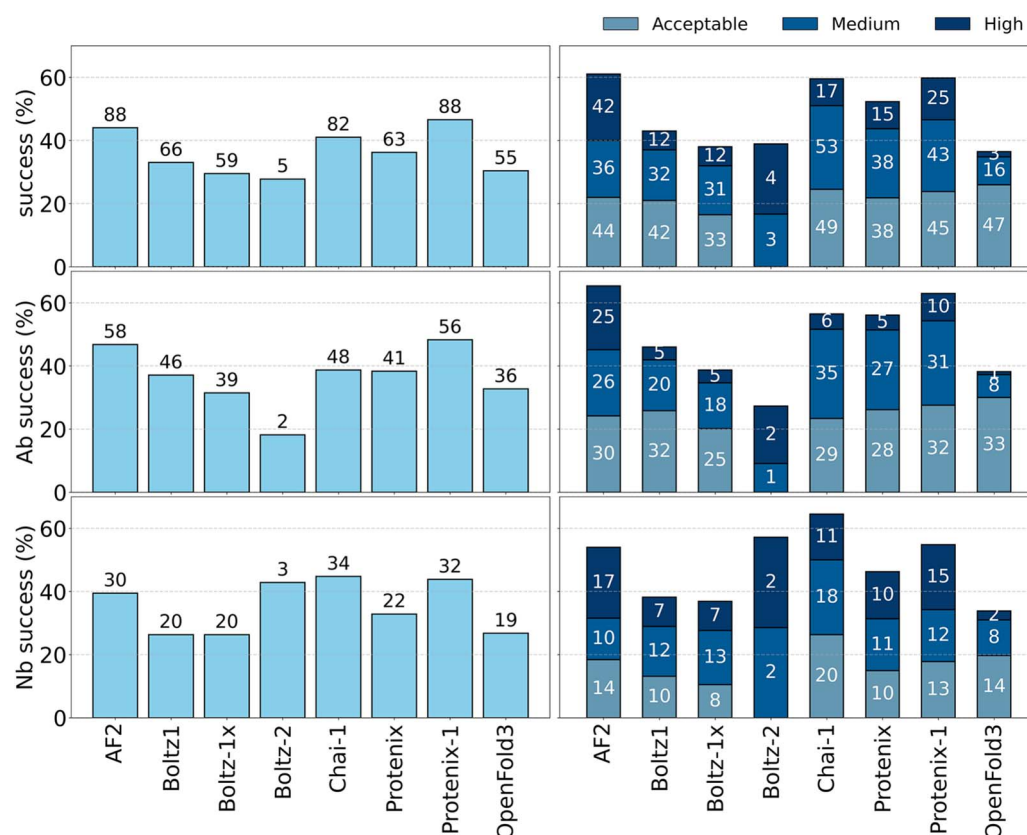


Figure 1 Success rates for (top row) all complexes (total, 200; Boltz-2, 18; Protenix, 174; Protenix-1, 191; OpenFold3, 181), (middle row) Ab complexes (Abs, 124; Boltz-2, 11; Protenix, 107; Protenix-1, 117; OpenFold3, 71) and (bottom row) Nb complexes (Nbs, 76; Boltz-2, 7; Protenix, 67; Protenix-1, 74; OpenFold3, 110). Left column displays percentage of unique targets with at least acceptable accuracy. Right column shows the percentages within each correct CAPRI category.

Contrary to prior studies that reported superior performance of AF3 over AF2 for Ab–Ag predictions [8–11], we observed comparable or higher performance of AF2 relative to the AF3-based implementations. We note that our AF2 predictions involved sampling 25 models from 5 seeds, while all AF3-based implementations generated 25 models from a single seed. This difference in sampling density likely explains the discrepancy, as it provides AF2 with a broader exploration of the conformational landscape compared with the single-seed inference used by AF3-based methods. Importantly, this observation also highlights the robustness of Protenix-1 and Chai-1, as they performed competitively against AF2 despite this single-seed constraint.

ESMFold failed to produce any successful predictions for our Ab–Ag complex set, as evaluated by both CAPRI and DockQ criteria (Fig. S2). Although this outcome is not entirely unexpected due to previous reports on the relatively poor performance for ESMFold for multimer predictions [27, 29], we repeated ESMFold predictions on a small subset using an different platform (<https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/ESMFold.ipynb>) to rule out implementation-related issues. Results from both predictions were identical in terms of complex structures and confidence scores. Therefore, we reported that while ESMFold has been reported to be highly accurate for monomeric structures [23], it failed to predict any of the complexes in this Ab–Ag set.

Previous benchmarking of AF2 v2.3.0 on a subset of our dataset showed ~43% overall success for top-25 models across individual CAPRI levels [5]. In our current benchmarking, AF2 v2.3.2 achieved

over 60% success (Fig. 1, right), which was close to the success of AFsample [30] reported in the same study [5]. These observed differences between the accuracies of two AF2 v2.3 versions could stem from variations in parameters, such as random seed initialization. Additionally, our use of ColabFold, which differs from the AF2 pipeline in its construction and pairing of MSAs [20], might also contribute to this discrepancy.

For Ab complexes (Fig. 1, middle row), Protenix-1 and AF2 emerged as the most successful methods, both achieving a success rate of 47%–48%. Following these, Chai-1 reached a 39% success rate, closely followed by Protenix at 38%. In the Nb level assessment (Fig. 1, bottom row), Chai-1 yielded the highest success rate (45%), outperforming Protenix-1 (43%), and AF2 (39%). Notably, most other methods showed lower performance on Nb complexes compared with their results on Ab complexes. Despite the limited tested complexes (18), Boltz-2 showed a higher success in Nb category, correctly predicting 3 out of 7 targets. Again, AF2 remained the superior method for producing the largest number of high-quality models for both Ab and Nb complexes, with Protenix-1 following closely behind. Notably, the newer Boltz versions (Boltz-1x and Boltz-2) showed lower performance on Ab complexes than the original Boltz-1, though they performed comparably or better on Nb complexes. In contrast, Protenix-1 demonstrated a notable improvement over its predecessor on both Ab and Nb complexes, consistently showing the highest success rate (Fig. 1).

We also evaluated all methods on the 18 complexes deposited after Boltz-2's training cutoff. Success rates were consistently lower for

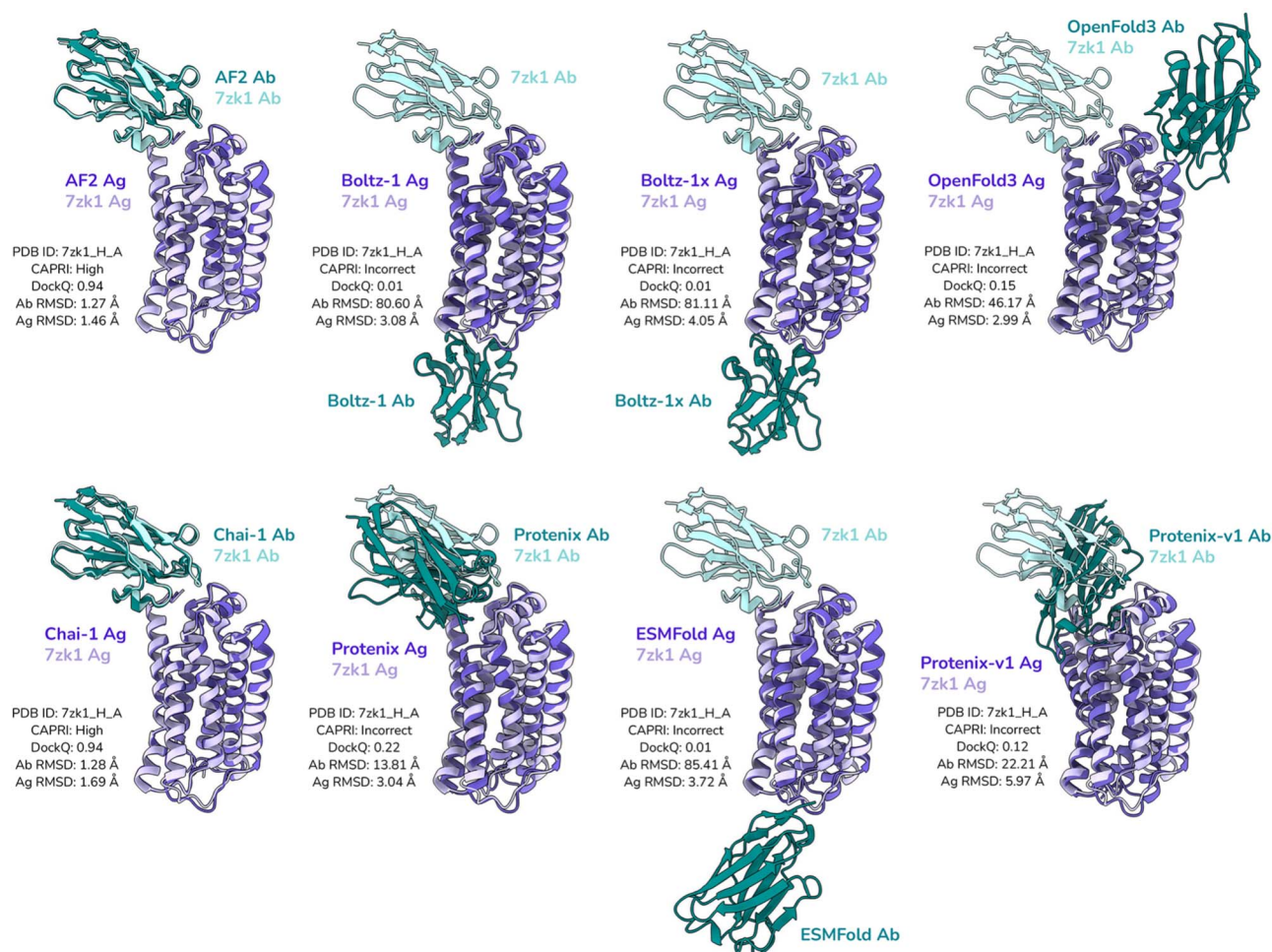


Figure 2 Structural visualization of 7zk1 complex showing the top-ranking models. Native structure is depicted in lighter colors, predicted models in darker shades. C α RMSD was given for each subunit.

this subset across all methods (Fig. S4), suggesting that these more recent complexes are particularly challenging. Notably, only AF2 and Protenix surpassed Boltz-2's performance on this subset, while all other methods performed below Boltz-2.

A recent study utilizing 70 Nb complexes reported a 33% success rate for AF2 v2.3.2 [10]. In our benchmark of 76 Nb complexes, AF2 v2.3.2 correctly predicted 30 targets, corresponding to a 39% success rate (Fig. 1, bottom), and produced a higher proportion of high-quality models (24%, 17/76) compared with the 11% reported in [10]. These differences in AF2 performance likely stem from variations in calculation parameters such as the number of recycling iterations, as well as the fact that the earlier benchmark used DockQ thresholds as ground truth, whereas we applied CAPRI criteria. The same study reported a 47% success rate for AF3 on Nb complexes, which is slightly above the 45% and 43% success rates we observed for Chai-1 and Protenix-1, respectively (Fig. 1, bottom). Eshak and Goupil-Lamy [10] also showed a higher percentage of high- and medium-quality models for AF3 predictions of Nb complexes than those reported here for the top performers, Chai-1 and Protenix-1 (Fig. 1, bottom). Although our benchmark and the prior Nb-focused study [10] used different ground-truth criteria that do not perfectly agree (Fig. S1), our results suggested that Chai-1 is the best predictor for Nb complexes. The top-scoring models from all methods were visualized for a representative

Nb complex (Fig. 2), and consistent with the overall Nb success rates (Fig. 1, bottom), Chai-1 achieved the most accurate prediction with the best scores overall.

In summary, among tested nine methods, Protenix-1 emerged as the best predictor for this Ab–Ag complex set, with Chai-1 and AF2 closely trailing in performance. AF2 was particularly consistent in generating high-quality complexes, while Chai-1 stood out as the top method for Nb modeling. Boltz variants generally showed lower and more variable performance. Notably, despite being tested on a limited number of complexes, Boltz-2 performed well particularly in generating high-quality models and in Nb modeling.

Effect of interface glycans on success rates

Glycosylation is a key post-translational modification (PTM) for both Abs and Ags, influencing their interaction [31]. In our dataset, we identified 41 Ab–Ag interfaces containing glycan residues (within 10 Å of the interface). While AF3-based methods can account for PTM/glycosylations during modeling, we did not provide any glycan information during inference. Hence, to assess the performance in the glycan-free versus glycosylated interfaces, we calculated the success rates for the complexes with and without interface glycans (Fig. S5). This analysis revealed a general decline in success for the complexes

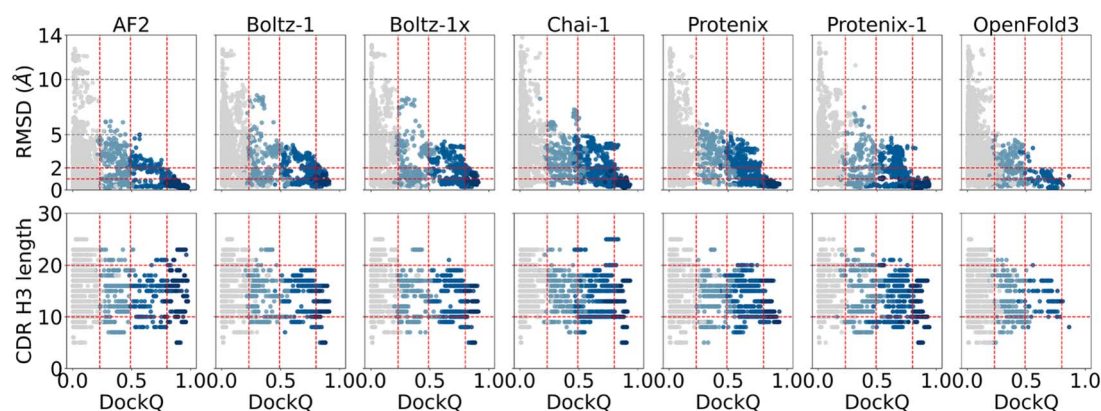


Figure 3 (Top) $C\alpha$ RMSD of the CDR-H3 loop is plotted against DockQ. Horizontal lines mark 1 and 2 Å. (Bottom) Length of the CDR-H3 loop is plotted against DockQ. In both panels, points are classified according to their CAPRI system, and vertical lines delineate DockQ thresholds for acceptable (0.23), medium- (0.49), and high-quality (0.80) models.

with glycosylated interfaces, relative to those lacking interface glycans. In the absence of explicit glycan input, OpenFold3 predicted interface glycans in only five complexes, producing only acceptable quality structures, implying a marked sensitivity of OpenFold3 to modified binding interfaces. Notably, Protenix variants stood out as the least affected methods, maintaining consistent success rates for both interfaces. Despite a lower success than Protenix variants, AF2 and Chai-1 produced a higher proportion of high-quality models for glycosylated interfaces. In contrast, Boltz-1 variants failed to produce any high-quality predictions for glycosylated interfaces. Overall, these results suggest that Boltz-1 variants and OpenFold3 are more sensitive to interface glycans than other methods, and that incorporating PTM information during inference could improve their performance.

We visualized a specific complex of an Nb and SARS-CoV-2 RBD (PDB ID: 7z2l) featuring an N-linked glycosylation at the 434th position of the RBD (Fig. S6). Even without explicit glycosylation input, AF2, Chai-1, and Protenix successfully produced medium- or high-quality models, whereas Boltz-1 variants failed. In contrast, we also identified three other complexes (PDB IDs: 7wg3, 8ce4, and 8ido) where all methods failed (Fig. S7). Their structural inspection revealed distinct patterns such as a glycan positioned at the center of interface (7wg3), two glycans flanking the interface (8ce4), and a large branched glycan structure at the interface (8ido). The failure of these targets suggested that while AF3-inspired methods except for OpenFold3 can occasionally succeed without explicit glycan input, inclusion of PTMs might be essential for higher accuracy particularly when glycans comprise a large fraction of the interface.

Effect of depth, recycling iterations, and seed

We calculated success rates for the models in top-1 and top-5 ranking tiers. Figure S8 displays the sum of target counts for each individual CAPRI category. The depth of prediction, meaning the number of predicted models, positively impacted the performance of all methods, resulting in higher success rates. The original AF3 method previously achieved a 60% success rate in Ab–Ag complex prediction using a 1000-seed sampling strategy [7]. However, a recent study reported a significantly lower success rate of ~11% with single-seed sampling [8], underscoring the sensitivity of the method to sampling depth. Our findings similarly indicated that AF2 and all AF3-based methods

responded well to an increased number of models in a given prediction (Fig. S8).

Recycling, which iteratively refines the model by reintroducing the outputs into the network for further optimization [4, 6], is an important factor contributing to the success of Ab–Ag modeling [5]. To assess the impact of recycling on model performance, we compared predictions generated with 3 recycling iterations to those generated with 20 (Fig. S8). Increasing recycling iterations generally improved performance across all methods, with Chai-1 and Protenix-1 showing the most pronounced gains. Specifically, correct predictions from Chai-1 rose from 45 to 67 for top-1 ranked models, from 50 to 70 for the top-5 tier, and from 70 to 82 across all 25 models (Fig. S8, top). Protenix-1 and Protenix similarly benefited from additional recycling iterations, with improvements observed across all tiers. AF2 and Boltz-1 also demonstrated consistent, albeit more modest, gains in prediction accuracy. In contrast, Boltz-1x exhibited a slight decline in success rates, particularly at the top-1 and top-25 tiers. Collectively, these results indicated that while increased recycling generally enhances accuracy in both AF2- and AF3-based methods; yet its effect is not universal and may even impair performance, as observed for Boltz-1x. Finally, to assess the influence of random seed variability on AF2 performance, an additional prediction was conducted using 3-recycle iterations with a different seed (Fig. S8, AF2 (3)*). Although overall success rates remained comparable between the two 3-recycle runs, the distribution of CAPRI quality classes shifted with different seed selection, suggesting that random seed choice can affect AF2's prediction quality independently of accuracy.

Structural fidelity of CDR-H3 loop against model quality

Modeling CDR-H3 loop is a central challenge in computational antibody design due to its high sequence and structural diversity [3, 32, 33]. To assess the structural accuracy of the predicted CDR-H3 loops, we computed its root mean square deviation (RMSD) relative to the native conformation (Fig. 3, top). For high-quality models, every method captured CDR-H3 conformations within 2 Å of the native conformation (Fig. 3, top). Protenix-1, AF2, Chai-1, and Protenix produced a large fraction of high-quality models with RMSDs below 1 Å, while Boltz-1 variants generally produced targets with CDR-H3

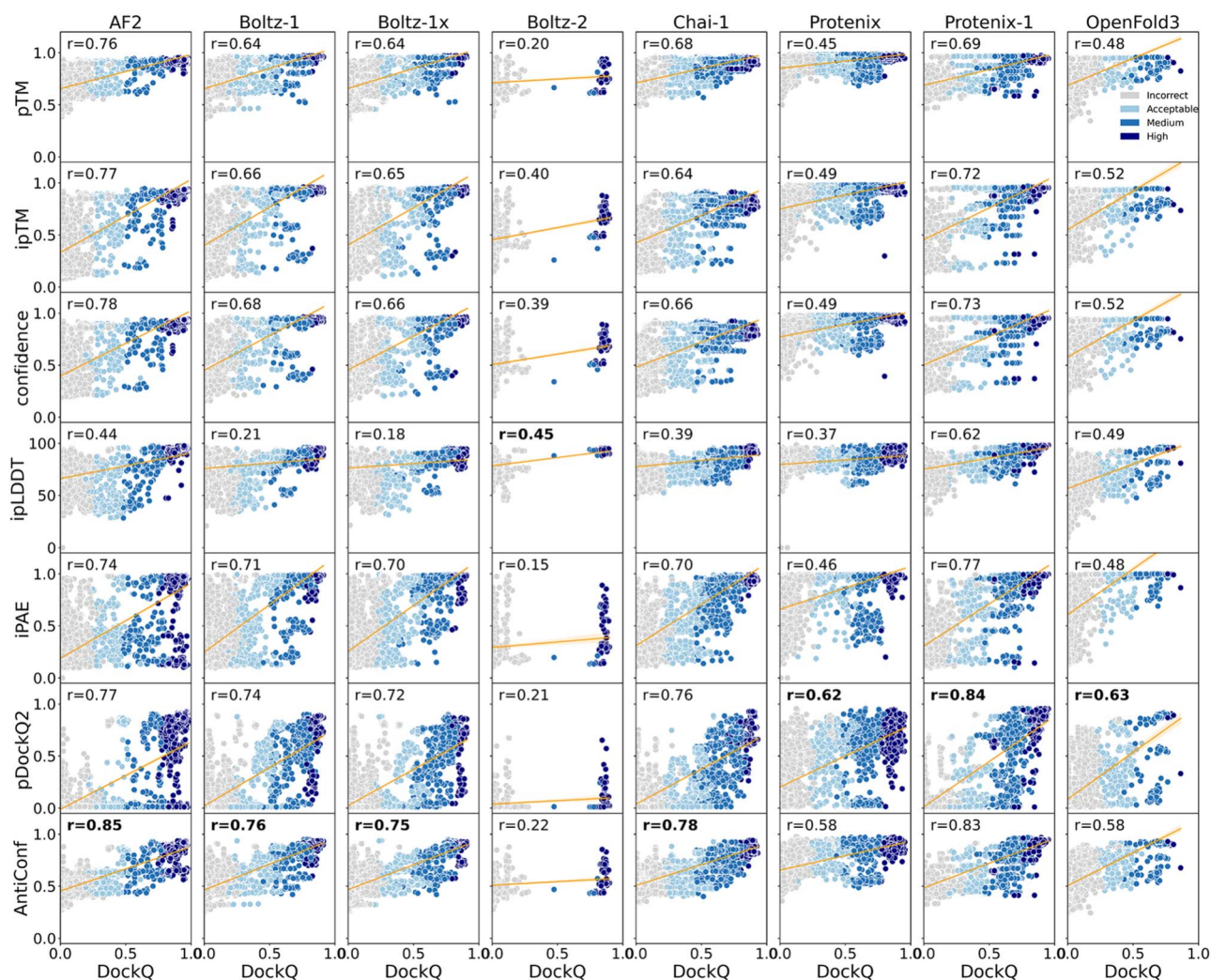


Figure 4 Scatter plots of DockQ scores and confidence metrics across eight structure prediction methods. All correlations are significant ($P < .001$). ipLDDT, iPAE, and pDockQ2 scores for obtained for the antigen chain only.

loop accuracy in the 1–2 Å range. Medium-quality models from all methods stayed below 5 Å, with OpenFold3 and AF2 standing out, as almost all of their medium-quality models fell below 2 and 3 Å, respectively. Nearly all acceptable-quality models from AF2, Chai-1, and Protenix showed CDR-H3 RMSDs under 5 Å, whereas Boltz-1 variants exhibited slightly elevated values in their acceptable-quality predictions. CDR-H3 RMSD values were right-skewed, ranging from 0 to 14 Å for all methods. Overall, this analysis showed that CDR-H3 loop accuracy is strongly correlated with model quality for all methods, reflecting the promise of Protenix-1, AF2, Chai-1, and Protenix for capturing native CDR-H3 loop conformations. Eshak and Goupil-Lamy [10] and Hitawala and Gray [19], which used a dataset overlapping with ours, reported comparable CDR-H3 RMSD distributions across DockQ scores, for AF3 predictions, implying a consistency for the CDR-H3 accuracy between the original AF3 and AF3-based methods tested here.

We also examined the impact of CDR-H3 loop length on model quality (Fig. 3, bottom). AF2 and Chai-1 consistently produced high- or medium-quality models across all loop lengths. Protenix performed best on medium-length loops, with few successes for loops shorter than 10 aa or longer than 20 aa, while Protenix-1 showed a compar-

tively weaker length bias, successfully predicting a higher proportion of long loops (> 20 aa). Boltz-1 variants similarly performed well on shorter loops while struggled with longer CDR-H3 loops. Notably, Chai-1 performed well on both short (<10 aa) and long loops (>20 aa), being the only method to correctly model the longest CDR-H3 loop in the 7mzm structure (25 aa). All other methods failed on this complex. Overall, these findings identified Chai-1 as the most successful method regardless of CDR-H3 length.

Evaluation of confidence scores as predictors of structural accuracy

To evaluate how well model confidence scores reflect structural accuracy, we analyzed the relationship between DockQ scores and confidence scores (Fig. 4). Incorrect models frequently received high confidence scores, underscoring the risk of false positives across the evaluated metrics. While pDockQ2 effectively assigned low scores to incorrect models, it did so at the cost of underrating correct ones, leaving many medium- and high-quality predictions with near-zero values. A similar tendency was observed for iPAE, which scored both high- and medium-quality models poorly. Since pDockQ2 is derived

by fitting a sigmoid curve to the product of antigen ipLDDT and iPAE scores [27], its failure to reward correct models can be largely attributed to the iPAE component (Fig. 4). In contrast, scores such as pTM, ipTM, model confidence, and ipLDDT consistently ranked correct models highly, with pTM providing the clearest separation of true positives in the high-score range. Given these complementary failure modes, false positives associated with pTM and false negatives with pDockQ2, we sought to combine their strengths into a single metric. To this end, we developed a composite score termed AntiConf, defined as a weighted average of pTM and pDockQ2, designed to balance sensitivity and specificity in Ab–Ag complex quality assessment.

Among the confidence scores, the strongest correlations were consistently found for pDockQ2 and AntiConf whilst ipLDDT had the weakest correlations, except for Boltz-2, which was represented by much fewer data points than other methods (Fig. 4). Among the tested methods, AF2 scores showed the strongest DockQ correlations, followed by Protenix-1 and Chai-1 scores. Notably, average ipLDDT computed for the entire interface correlated more strongly with DockQ compared with antigen-only ipLDDT for all methods (Fig. S9a), suggesting that averaging ipLDDT over the whole complex yields a more robust indicator of model accuracy for this Ab–Ag complex set. Unlike this trend seen with ipLDDT, antigen-specific iPAE showed higher correlations than full-interface iPAE scores (Fig. S9b).

The ipTM is a confidence score used to evaluate protein–protein interactions by assessing the pTM scores of interface contacts [34]. Alternative scores for interchain contacts have been developed [35, 36]. Among these, we included actifpTM scores for evaluation of AF2 predictions (Fig. S10). ActifpTM, which considers the residues directly involved in the interaction, showed a weaker association with DockQ compared with ipTM, largely due to overrating a subset of incorrect models.

Examining the distributions of model scores across predictions with varying recycling iterations, we observed an upward trend in nearly all confidence scores as the number of recycles increased, with the scores from AF2 and Protenix-1 being the least responsive to this effect for both incorrect (Fig. S11) and correct predictions (Fig. S12). Notably, Protenix scores were systematically overestimated relative to all other methods, for both correct and incorrect predictions. This overestimation was largely corrected in the updated version, Protenix-1, reflecting an improvement not only in its accuracy (Fig. 1) but also in the calibration of confidence scores.

Binary classification performance of confidence scores

Binary classification performance of confidence scores was evaluated using CAPRI criteria as ground truth (Table S1). We selected a threshold of 0.80 for all scores, which is a recognized threshold for ipTM in EMBL-EBI guidelines and prior studies for high-confidence interfaces [37–39]. AntiConf and pDockQ2 were the two scores that yielded the highest precision across all methods. Despite their comparable strength in precision, AntiConf consistently showed higher recall than pDockQ2 for all methods, resulting in higher F1 score (Table S1). In line with this, the proportion of successfully predicted unique targets, termed $\text{recall}_{\text{target}}$, was also notably higher for AntiConf than it was for pDockQ2, reflecting the potential of AntiConf in recalling a higher number of targets without compromising precision (Table S1). While both scores achieved precision above 0.9 for Protenix-1, Chai-1, and

AF2 at this threshold, AntiConf recalled 39 targets for each of Protenix-1 and AF2 and 37 targets for Chai-1, while this number was reduced to 26 for pDockQ2 for Protenix-1 and AF2 and 9 targets for Chai-1. These results collectively highlight a more robust precision–recall balance for AntiConf relative to pDockQ2.

While the high recall power of pTM, ipTM, and model confidence was also recognized (Table S1), high precision is arguably more critical in real-world applications where experimental structures are unavailable. Thus, AntiConf, by integrating the recall strength of pTM with the precision of pDockQ2, offered a compelling solution to the reduced recall associated with pDockQ2 alone. AntiConf thus offers a practical advantage in real-world applications where experimental structures are unavailable and misclassifying incorrect structures could lead to superfluous experimental follow-ups.

To further test the robustness of the scores under stringent precision conditions, we found the lowest threshold (T) that achieved a precision of 1, which are listed in Table 1. In this specific scenario, guaranteeing a precision of 1, AntiConf either performed the best or the second in terms of recall and/or $\text{recall}_{\text{target}}$ (Table 1). For instance, for AF2, AntiConf yielded the second-highest recall but ultimately identified the highest number of unique targets (27) (Table 1). We then progressively decreased T by 0.05, 0.10, and 0.15 and evaluated performance at these progressively decremented thresholds for each method (Figs S13–S19). As anticipated, a monotonically increased recall performance was observed with decreasing thresholds. However, a concurrent decrease in precision was noted at decreasing thresholds for all scores, except for pDockQ2 and AntiConf, which maintained a high level of precision even at $T = 0.15$. Specifically, AntiConf achieved a precision of 0.96 and recalled 42 unique targets for AF2, whereas pDockQ2 recalled 33 targets with a precision of 0.95 (Fig. S13). In contrast, the same condition led to a reduced precision (0.59–0.86) for other scores (Fig. S13).

For the AF3-based implementations, the same pattern held. AntiConf consistently maintained a high level of precision across the decremented thresholds, while its recall and $\text{recall}_{\text{target}}$ were always higher than those of the second-best precision performer, pDockQ2 (Figs S14–S19). For Chai-1 and Protenix-1, AntiConf's precision was slightly lower than that of pDockQ2, specifically at thresholds decremented by 0.10 and 0.15 units (Figs S16 and S18). The largest difference between the two metrics was observed at $T = 0.15$ for both Chai-1 and Protenix-1, where pDockQ2 achieved precisions of 0.92 and 0.96, compared with 0.88 and 0.90 for AntiConf. Despite these small compromises in precision, AntiConf showed substantially higher recall and $\text{recall}_{\text{target}}$, identifying 48 targets compared with pDockQ2's 30 for Chai-1 (Fig. S16), and 42 targets compared with 29 for Protenix-1 (Fig. S18). For both Boltz-1 variants, AntiConf achieved the highest precision and recall across all thresholds, including the decremented ones (Figs S14 and S15). Although evaluated on a small dataset, we also report the performance of Boltz-2 scores across varying thresholds (Table S2). Notably, ipLDDT achieved the highest performance at every threshold, followed by pTM, while other metrics, including AntiConf, showed no particular advantage for Boltz-2 predictions.

For Protenix and OpenFold3, AntiConf's robustness against threshold decrements was less clear, as it showed a larger precision loss across all tested thresholds than pDockQ2 (Figs S17 and S19). Nonetheless, AntiConf's recall advantage was still noticeable. For instance, at $T = 0.15$ in Protenix, AntiConf yielded a precision of 0.67 compared with pDockQ2's 0.81, but with a much larger gain in recall and $\text{recall}_{\text{target}}$, identifying 42 targets compared with pDockQ2's

Table 1 Classification metrics based on the lowest score threshold at precision 1

Method*	Score	Threshold	Recall	F1	Targets	Recall _{target}
AF2 (88)	pTM	0.935	0.147	0.257	16	0.182
	ipTM	0.916	0.168	0.288	21	0.239
	modelconf.	0.919	0.159	0.274	20	0.227
	iPAE	0.98	0.032	0.061	4	0.045
	ipLDDT	0.952	0.244	0.392	27	0.307
	pDockQ2	0.848	0.152	0.264	19	0.216
	AntiConf	0.879	0.235	0.381	27	0.307
Boltz-1 (66)	pTM	0.969	0.036	0.07	3	0.045
	ipTM	0.96	0.02	0.039	2	0.03
	modelconf.	0.961	0.025	0.049	2	0.03
	iPAE	0.991	0.039	0.076	3	0.045
	ipLDDT	0.964	0	0	0	0
	pDockQ2	0.899	0.001	0.002	1	0.015
	AntiConf	0.914	0.085	0.157	4	0.061
Boltz-1x (59)	pTM	0.968	0.051	0.097	5	0.085
	ipTM	0.958	0.034	0.066	3	0.051
	modelconf.	0.96	0.038	0.073	4	0.068
	iPAE	0.992	0.039	0.075	2	0.034
	ipLDDT	0.957	0.001	0.002	1	0.017
	pDockQ2	0.897	0.001	0.002	1	0.017
	AntiConf	0.913	0.082	0.152	4	0.068
Chai-1 (82)	pTM	0.962	0	0	0	0
	ipTM	0.919	0	0	0	0
	modelconf.	0.928	0	0	0	0
	iPAE	0.956	0.223	0.365	21	0.256
	ipLDDT	0.968	0.017	0.034	1	0.012
	pDockQ2	0.753	0.141	0.247	13	0.159
	AntiConf	0.874	0.159	0.274	13	0.159
Protenix (63)	pTM	0.985	0.056	0.105	6	0.095
	ipTM	0.982	0	0	0	0
	modelconf.	0.982	0.001	0.002	1	0.016
	iPAE	0.996	0.005	0.009	1	0.016
	ipLDDT	0.985	0	0	0	0
	pDockQ2	0.955	0.001	0.002	1	0.016
	AntiConf	0.976	0.001	0.002	1	0.016
Protenix-1 (88)	pTM	0.965	0.1	0.182	6	0.068
	ipTM	0.951	0.272	0.427	17	0.193
	modelconf	0.954	0.211	0.348	13	0.148
	iPAE	0.991	0.126	0.224	9	0.102
	ipLDDT	0.959	0.229	0.373	16	0.182
	pDockQ2	0.899	0.186	0.313	12	0.136
	AntiConf	0.913	0.223	0.365	16	0.182
OpenFold3 (55)	pTM	0.954	0.031	0.06	1	0.018
	ipTM	0.943	0.023	0.045	1	0.018
	modelconf	0.945	0.023	0.045	1	0.018
	iPAE	0.998	0.046	0.088	1	0.018
	ipLDDT	0.934	0.098	0.178	2	0.036
	pDockQ2	0.859	0.098	0.178	2	0.036
	AntiConf	0.898	0.129	0.228	3	0.055

Numbers in parentheses indicate the count of correctly predicted unique targets for each method. For each method, the best-performing result is shown in bold.

25. Similarly, for OpenFold3, despite a drop in precision from 0.56 to 0.35, AntiConf surpassed pDockQ2 in recall by correctly identifying 20 targets compared with only 6 recalled by pDockQ2. Altogether, AntiConf remained the best metric for maintaining an optimal precision–recall balance, and even in cases where pDockQ2 performed better in terms of precision, AntiConf consistently achieved a higher recall/recall_{target} while retaining reasonable precision. It is important to note that the pDockQ2 metric was originally parameterized using AF2 scores from protein–protein complexes [40]. Thus, applying its AF2-calibrated parameters to AF3-based implementations and/or specific Ab–Ag complexes might introduce bias, espe-

cially considering the overestimation of iPAE in AF3-based methods (Fig. S11). Hence, a method-specific calibration might further ameliorate the predictive power of pDockQ2 and thus AntiConf. We evaluated another precision-stringent scenario that captures at least 95% precision for all scores (Table S3). If a score could not achieve precision ≥ 0.95 , we reported the threshold yielding the second highest precision (Fig. S17). This condition similarly reflected AntiConf's success for AF2, Boltz-1, and Chai-1 based on recall and recall_{target}. Altogether, these analyses showed that AntiConf tolerated threshold decrements robustly, maintaining a stable and high level of precision. Furthermore, it also showed a high recall

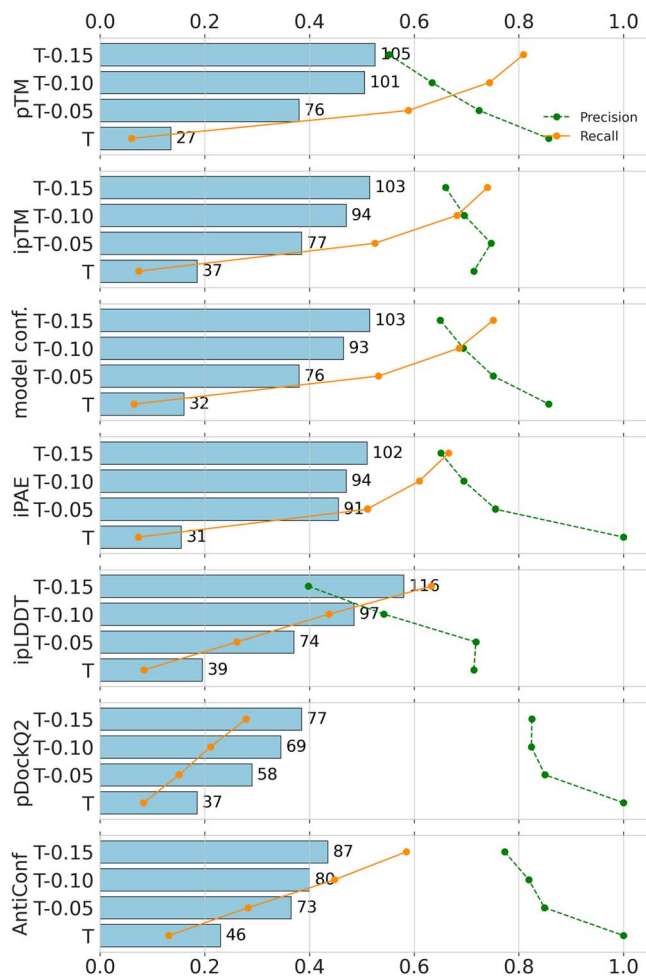


Figure 5 Collective precision, recall, and recall_{target} performance of all methods based on thresholds starting from the one that gives precision 1 (T) and decrementing it by 0.05, 0.10, and 0.15. See Figs S13–S19 for corresponding values of T for each method and score.

and recall_{target}, whilst no other scores displayed a comparable performance.

We lastly evaluated the precision–recall performance of the confidence scores by concatenating all positive predictions from all methods. We conducted this analysis for the threshold that yield precision of 1 and three incrementally lower thresholds. Using these thresholds, we combined the positives predicted by each method into a single set of positive and obtained performance metrics for each score (Fig. 5). Consistent with the assessments based on individual methods, when predictions from all methods were combined, pDockQ2 and AntiConf delivered the highest recall and recall_{target}. They consistently showed increased recall and recall_{target} while maintaining high precision. Notably, AntiConf consistently demonstrated the highest recall_{target} across all tested thresholds, highlighting its potential for evaluating future Ab–Ag complexes predicted by AF2 or AF3-based implementations.

Conclusion

Our study provides a comparative assessment of publicly available AF2 and AF3-based methods, highlighting the strengths of Protenix-1 AF2 and Chai-1 in accurate determination of Ab–Ag complexes.

We introduced AntiConf, a novel metric optimized for maximizing recall within high-precision scenarios. By effectively distinguishing correct Ab–Ag complexes from AF2 and AF3 predictions, AntiConf serves as a robust filtering strategy for antibody design, bridging the gap between sequence generation and high-quality structural modeling. We propose AntiConf as a gate score for both computational and experimental workflows, where method-specific thresholds can be tuned to achieve the desired precision–recall trade-off, ultimately facilitating the reliable selection of epitope-specific binders.

Our work also shows that predictions from multiple methods can be combined to maximize prediction accuracy. Combining different prediction methods, rather than relying on a single methodology, it is possible to extract a greater number of correct predictions. Collectively, our benchmarking supports a pragmatic, multi-model strategy for future end-to-end modeling pipelines beginning with Ab sequences.

Key Points

- **Comprehensive benchmarking:** For antibody–antigen complexes, Protenix-1 was the strongest predictor, with Chai-1 as a close second and specific strengths on nanobody complexes. AF2 remained a strong method for capturing high/medium-quality interfaces.
- **Method performance insights:** Additional recycling improves AF2, Chai-1, and Protenix variants but not Boltz-1 variants, and generating more models generally boosts accuracy. Glycosylation proves essential, with the Protenix methods the most resilient, and Chai-1 best capture native CDR-H3 loops across different lengths.
- **Confidence-score evaluation and antibody confidence (AntiConf):** Existing confidence scores often misestimate structural quality. Our composite AntiConf score achieves the strongest correlation with model accuracy.
- **Precision-focused guidelines:** When tested under strict precision constraints, AntiConf outperforms all other metrics, maximizing recall in high-precision scenarios. A comprehensive threshold recommendations were finally provided for future evaluation studies.

Supplementary material

Supplementary material is available at *Briefings in Bioinformatics* online.

Conflicts of interest

S.U. and B.H. are affiliated with and E.T. serves as a scientific consultant for the Antiverse Ltd. I.S. declares no competing interests.

Data availability

Dataset, predictions, and calculated scores are available via Zenodo (<https://doi.org/10.5281/zenodo.16404871>). The scripts for calculation of confidence scores are available via GitHub (<https://github.com/timucinlab/Ab-Ag-prediction>).

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